		Data Type xsd:string
D_337 Time	Time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99)	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 6 **max:** 6 **External Category** Element **Data Type** xsd:string
D_28 Group Control Number	Assigned number originated and maintained by the sender	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 9 **max:** 9 **External Category** Element **Data Type** xsd:string
D_455 Responsible Agency Code	Code identifying the issuer of the standard; this code is used in conjunction with Data Element 480 **Selected Values** All Values in Codelist	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 1 **max:** 1 **External Category** Element **Data Type** xsd:string **Codelist Id** D_455 **Type System** Customer_TS **Version Mode** Local **Version** 1.1
D_480 Version / Release / Industry Identifier Code	Code indicating the version, release, subrelease, and industry identifier of the EDI standard being used, including the GS and GE segments; if code in DE455 in GS segment is X, then in DE 480 positions 1-3 are the version number; positions 4-6 are the release and subrelease, level of the version; and positions 7-12 are the industry or trade association identifiers (optionally assigned by user); if code in DE455 in GS segment is T, then other formats are allowed **Selected Values** All Values in Codelist	**Cardinality** **min:** 1 **max:** 1 **Primitive Type** String **Length** **min:** 6 **max:** 6 **External Category** Element **Data Type** xsd:string **Codelist Id** D_480 **Type System** Customer_TS **Version Mode** Local **Version** 1.1

2.5 S_GE — Functional Group Trailer

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
├ **FunctionalGroup** — Functional Group	1 .. 1				
│ └ **S_GE** — Functional Group Trailer	1 .. 1				
│ ├ **D_97** — Number of Transaction Sets Included	1 .. 1	String		1 .. 1	
│ └ **D_28** — Group Control Number	1 .. 1	String		9 .. 9	

Documentation

Definition To indicate the end of a functional group and to provide control information Comments 1. The use of identical data interchange control numbers in the associated functional group header and trailer is designed to maximize functional group integrity. The control number is the same as that used in the corresponding header. Semantic Notes: 1. The data interchange control number GE02 in this trailer must be identical to the same data element in the associated functional group header, GS06.

Properties

Identifier S_GE

Name Functional Group Trailer

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_97 Number of Transaction Sets Included	Total number of transaction sets included in the functional group or interchange (transmission) group terminated by the trailer containing this data element	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string
D_28 Group Control Number	Assigned number originated and maintained by the sender	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 9 max: 9 **External Category** Element **Data Type** xsd:string

2.6 S_IEA — Interchange Control Trailer

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
Interchange — Interchange Structure	1 .. 1				
└ **S_IEA** — Interchange Control Trailer	1 .. 1				
├ **D_I16** — Number of Included Functional Groups	1 .. 1	String		1 .. 1	
└ **D_I12** — Interchange Control Number	1 .. 1	String		9 .. 9	

Documentation

Definition To define the end of an interchange of zero or more functional groups and interchange-related control segments

Properties

Identifier S_IEA

Name Interchange Control Trailer

Cardinality min: 1 max: 1

Syntax Type Related

External Category Element

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
D_I16 Number of Included Functional Groups	A count of the number of functional groups included in an interchange	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 1 **External Category** Element **Data Type** xsd:string
D_I12 Interchange Control Number	A control number assigned by the interchange sender	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 9 max: 9 **External Category** Element **Data Type** xsd:string

3 OVERVIEW MESSAGE

General Information

Name	SAP_LBN_ANSI_X12_840
Direction	Out
Status	Active
Message Type	Request for Quotation
Type System	ASC_X12
Version	004010
External Category	Transaction Set

Documentation

Definition This Draft Standard for Trial Use contains the format and establishes the data contents of the Request for Quotation Transaction Set (840) for use within the context of an Electronic Data Interchange (EDI) environment. The transaction set can be used to provide potential buyers with the ability to solicit price, delivery schedule, and other items from potential sellers of goods and services.

Notes

Usage 1 RFQ Transaction Type
Create RFQ: BQT*00*...~
Cancel RFQ: BQT*01*...~

Usage 2 RFQ notes are transmitted using a combination of N9 and MSG segments.

The notes section begins with an N9 segment using the “L1” qualifier:
N9*L1*NOTE_TYPE_TEXT~

Following the N9 header, each note is encoded in one or more MSG segments:
MSG*TYPE={ note_type} |TEXT={ note_content} ~

Notes Example:
N9*L1*NOTE_TYPE_TEXT~
MSG*TYPE=VSPIN|TEXT=Temperature controlled shipment required. Maintain 34-38 degrees F throughout transit. Reefer unit must be pre-cooled before loading.~
MSG*TYPE=TRUCK|TEXT=Truck Equipment Required: 53ft dry van with liftgate. Driver must have HAZMAT certification and valid TWIC card.~
MSG*TYPE=RMARK|TEXT=Preferred routing via I-95 corridor. Avoid congested urban areas during peak hours. Contact dispatcher 2 hours prior to delivery for dock assignment.~

Usage 3 Each transportation stage begins with an N9 segment using the “55” qualifier.
N9*55*{ stage_id} *{ stage_name} ~

Following the N9 segment, multiple DTM segments provide timing information for the stage.

After the DTM segments, a MSG segment encodes the distance information:
MSG*DISTANCE={ value} |UNIT={ unit_code} ~

Each transportation stage’s loadingLocation and unloadingLocation are mapped to N1*SF and N1*ST respectively, where REF*55*{ stage_id} is used to link to the specific transportation stage.

Transportation Stages Example

N9*55*1*STAGE1~
DTM*010*20260501*230000*UT~
DTM*017*20260518*090000*UT~
MSG*DISTANCE=180.03|UNIT=DK~

N9*55*2*STAGE2~
DTM*010*20260701*160000*UT~
DTM*017*20260715*170000*UT~
DTM*037*20260701*020000*UT~
DTM*038*20260701*100000*UT~
DTM*064*20260715*030000*UT~
DTM*063*20260715*190000*UT~
MSG*DISTANCE=2368.6|UNIT=DK~

N1*SF*ABC Logistics Center~
N3*123 Industrial Park Road~
N4*Frankfurt*HE*60486*DE~
REF*55*1*STAGE1~

N1*ST*XYZ Distribution Hub~
N3*456 Commerce Boulevard~
N4*Munich*BY*80331*DE~
REF*55*1*STAGE1~
PER*IC**TE*+49 89 1234567*EM*contact@xyz-hub.de~

N1*SF*Global Warehouse Solutions~
N3*789 Shipping Avenue*Building A~
N4*Hamburg*HH*20095*DE~
REF*55*2*STAGE2~
PER*IC**TE*+49 40 7654321*EM*info@global-warehouse.de~

N1*ST*Mediterranean Port Terminal~
N3*321 Harbor Street~
N4*Athens*AT*15123*GR~
REF*55*2*STAGE2~

Usage 4 The following control fields are transmitted using the N9 segment with qualifier code “ZZ” :
priceNegotiable
datesNegotiable
shippingTypeCode

N9*ZZ*PRICE_NEGOTIABLE*{ VALUE} ~
Allowed Values: Y and N
N9*ZZ*DATES_NEGOTIABLE*{ VALUE} ~
Allowed Values: Y and N
N9*ZZ*SHIPPING_TYPE*{ VALUE} ~
Allowed Values:
LTL
FTL
Parcel
Bulk
Loose
ULD
LCL
FCL

Usage 5 US Fluid Ounce (ISO code: OZA) is not supported by the ANSI X12 840 Volume Unit Qualifier. Therefore, when the source volume unit is OZA, it is mapped to LT (Liter). The volume value is converted from US Fluid Ounce to Liter using the standard conversion factor (1 US Fluid Ounce = 0.0295735295625 Liter) to preserve the original quantity.

Example 1 Create RFQ

ISA*00* *00* *ZZ*Shipper LBNID *ZZ*Carrier LBNID *260408*1130*U*00401*000003202*0*P*>~
GS*RQ*Shipper LBNID*Carrier LBNID*20260408*113000*000003202*X*004010~
ST*840*0001~
BQT*00*RFQ20260002*20260408~
CUR*BY*EUR~
REF*06*QW9_170~
REF*OQ*4500012346~
CTP***2500.00~
DTM*036*20260412*120000*UT~
MEA*PD*LN*520*DK~
MEA*WT*G*1800*KG~
TD5****M~
RRA*16~
N9*ZZ*PRICE_NEGOTIABLE*Y~
N9*ZZ*DATES_NEGOTIABLE*N~
N9*ZZ*SHIPPING_TYPE*FTL~
N9*L1*NOTE_TYPE_TEXT~
MSG*TYPE=RMARK|TEXT=Hazardous cargo included. Driver must be ADR qualified.~
N9*55*1*STAGE1~
DTM*010*20260415*080000*UT~
DTM*017*20260415*160000*UT~
MSG*DISTANCE=180|UNIT=DK~
N9*55*2*STAGE2~
DTM*010*20260416*070000*UT~
DTM*017*20260416*180000*UT~
MSG*DISTANCE=340|UNIT=DK~
N1*SF*Shipper Plant A~
N3*88 Logistics Road~
N4*Shanghai**200120*CN~
REF*55*1*STAGE1~
N1*ST*Transit Hub B~
N3*66 Hub Avenue~

N4*Nanjing**210000*CN~
REF*55*1*STAGE1~
N1*SF*Transit Hub B~
N3*66 Hub Avenue~
N4*Nanjing**210000*CN~
REF*55*2*STAGE2~
N1*ST*Consignee WH C~
N3*18 Industrial Ave~
N4*Suzhou**215000*CN~
REF*55*2*STAGE2~
PER*IC**TE*+86 512 12345678*EM*dock.suzhou@example.com~
N1*BY*LBN SHIPPER ORG~
N3*1 Shipper Plaza~
N4*Shanghai**200001*CN~
REF*CN*LBNBUYER01~
N1*CA*Preferred Carrier Ltd~
N3*99 Carrier Street~
N4*Shanghai**200122*CN~
REF*12*LBNCARR01~
PO1*1*8*DR~
MEA*WT*G*1200*KG~
PID*F****UN1993 Flammable liquid n.o.s.~
REF*55*1*STAGE1~
TD4*HM*D*3*UN1993, Flammable liquid, Class 3, PG II*Y~
PO1*2*4*DR~
MEA*WT*G*600*KG~
PID*F****UN1263 Paint related material~
REF*55*2*STAGE2~
TD4*HM*D*3*UN1263, Paint, Class 3, PG II*Y~
SE*60*0001~
GE*1*000003202~
IEA*1*000003202~

Example 2 Cancel RFQ

ISA*00* *00* *ZZ*Shipper LBNID *ZZ*Carrier LBNID *260317*0207*U*00401*000002930*0*P*>~
GS*RQ*Shipper LBNID*Carrier LBNID*20260317*020757*000002930*X*004010~
ST*840*0001~
BQT*01*RFQ20256013*20260317~
REF*06*QW9_170~
SE*4*0001~
GE*1*000002930~
IEA*1*000002930~

3.1 Structure

The following table shows only the root element and the group elements.

Node	Card.	Prim.Type	Pos.	Length	Codelists
840 — Request for Quotation	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
├ **BQT** — Beginning Segment for Request for Quotation	1 .. 1		020		
├ **CUR** — Currency	0 .. 1		040		
├ **REF** — Reference Identification	0 .. *		050		
├ **CTP** — Pricing Information	0 .. *		090		
├ **DTM** — Date/Time Reference	0 .. 10		150		
├ **MEA** — Measurements	0 .. 40		200		
│ └ **C001** — Composite Unit of Measure	0 .. 1		04		
├ **TD5** — Carrier Details (Routing Sequence/Transit Time)	0 .. 12		240		
├ **RRA** — Required Response	0 .. 100		280		
├ **N9** — Loop N9	0 .. 1000		290		
│ ├ **N9** — Reference Identification	1 .. 1		290		
│ ├ **DTM** — Date/Time Reference	0 .. *		295		
│ └ **MSG** — Message Text	0 .. 1000		300		
├ **N1** — Loop N1	0 .. 10000		310		
│ ├ **N1** — Name	1 .. 1		310		
│ ├ **N3** — Address Information	0 .. 2		330		
│ ├ **N4** — Geographic Location	0 .. *		340		
│ ├ **REF** — Reference Identification	0 .. 12		350		
│ └ **PER** — Administrative Communications Contact	0 .. *		360		
├ **PO1** — Loop PO1	1 .. 100000		010		
│ ├ **PO1** — Baseline Item Data	1 .. 1		010		
│ ├ **MEA** — Measurements	0 .. 40		049		
│ │ └ **C001** — Composite Unit of Measure	0 .. 1		04		
│ ├ **PID** — Loop PID	0 .. 1000		050		
│ │ └ **PID** — Product/Item Description	1 .. 1		050		
│ ├ **REF** — Reference Identification	0 .. *		100		
│ └ **TD4** — Carrier Details (Special Handling, or Hazardous Materials, or Both)	0 .. 5		270		
└ **SE** — Transaction Set Trailer	1 .. 1		020		

4 DETAILS MESSAGE

4.1 840 — Request for Quotation

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
840 — Request for Quotation	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
├ **BQT** — Beginning Segment for Request for Quotation	1 .. 1		020		
├ **CUR** — Currency	0 .. 1		040		
├ **REF** — Reference Identification	0 .. *		050		
├ **CTP** — Pricing Information	0 .. *		090		
├ **DTM** — Date/Time Reference	0 .. 10		150		
├ **MEA** — Measurements	0 .. 40		200		
├ **TD5** — Carrier Details (Routing Sequence/Transit Time)	0 .. 12		240		
├ **RRA** — Required Response	0 .. 100		280		
├ **N9** — Loop N9	0 .. 1000		290		
├ **N1** — Loop N1	0 .. 10000		310		
├ **PO1** — Loop PO1	1 .. 100000		010		
└ **SE** — Transaction Set Trailer	1 .. 1		020		

Properties

Identifier 840

Name Request for Quotation

Cardinality min: 1 max: 1

4.2 ST — Transaction Set Header

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
840 — Request for Quotation	1 .. 1				
├ **ST** — Transaction Set Header	1 .. 1		010		
│ ├ **143** — Transaction Set Identifier Code	1 .. 1	Token	01	3 .. 3	143
│ └ **329** — Transaction Set Control Number	1 .. 1	String	02	4 .. 9	

Documentation

Definition To indicate the start of a transaction set and to assign a control number

Properties

Identifier ST

Name Transaction Set Header

Cardinality min: 1 max: 1

Syntax Type Related

External Category Segment

Position 010

Level 2

Notes

Usage 1 The transaction set identifier (ST01) used by the translation routines of the interchange partners to select the appropriate transaction set definition (e.g., 810 selects the Invoice Transaction Set).

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
143 Transaction Set Identifier Code	Code uniquely identifying a Transaction Set **Selected Values** All Values in Codelist	**Cardinality** min: 1 max: 1 **Primitive Type** Token **Length** min: 3 max: 3 **External Category** Simple Data Element **Data Type** ID **Position** 01 **Codelist Id** 143 **Type System** ASC_X12 **Version Mode** Current **Version** 004010
329 Transaction Set Control Number	Identifying control number that must be unique within the transaction set functional group assigned by the originator for a transaction set	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 4 max: 9 **External Category** Simple Data Element **Data Type** AN **Position** 02

4.3 BQT — Beginning Segment for Request for Quotation

Structure

Node	Card.	Prim.Type	Pos.	Length	Codelists
840 — Request for Quotation	1 .. 1				
├ **BQT** — Beginning Segment for Request for Quotation	1 .. 1		020		
│ ├ **353** — Transaction Set Purpose Code	1 .. 1	Token	01	2 .. 2	353
│ ├ **586** — Request for Quote Reference Number	1 .. 1	String	02	1 .. 45	
│ └ **373** — Date	1 .. 1	Date	03	8 .. 8	

Documentation

Definition To indicate the beginning of a Request for Quotation Transaction Set and transmit identifying numbers and dates

Properties

Identifier BQT

Name Beginning Segment for Request for Quotation

Cardinality min: 1 max: 1

Syntax Type Related

External Category Segment

Position 020

Level 2

Notes

Usage 1 BQT03 is the date to be used for reference purposes in an RFQ and a response to RFQ.

Constraint 1 P0405 Paired Multiple - If any are used, all must be used

Leaf Elements

Identifier/Name	Description/Notes/Code Values	Properties
353 Transaction Set Purpose Code	Code identifying purpose of transaction set **Selected Values** **Code** 00 **Name** Original **Code** 01 **Name** Cancellation	**Cardinality** min: 1 max: 1 **Primitive Type** Token **Length** min: 2 max: 2 **External Category** Simple Data Element **Data Type** ID **Position** 01 **Codelist Id** 353 **Type System** ASC_X12 **Version Mode** Current **Version** 004010
586 Request for Quote Reference Number	Number assigned by the purchaser to identify his request for quote	**Cardinality** min: 1 max: 1 **Primitive Type** String **Length** min: 1 max: 45 **External Category** Simple Data Element **Data Type** AN **Position** 02
373 Date	Date expressed as CCYYMMDD	**Cardinality** min: 1 max: 1 **Primitive Type** Date **Length** min: 8 max: 8 **Date Time Format** CCYYMMDD **External Category** Simple Data Element **Data Type** DT **Position** 03